4 Fair Division

4.3 The Lone Divider Method

Before we dive into the Lone Divider Method, let's go over one more example of the Divider Chooser method.

Example 4.3.1 Suppose Jeff and Abed are splitting a Subway sandwich, which is half Veggie Delight and half chicken, and Jeff likes chicken 2 times as much as he likes veggies. Abed likes veggies 2 times as much as he likes Chicken. They use the divider chooser method to determine how to split the sandwich. The sandwich is worth \$16.

1. If Jeff is the divider, where does he cut the sandwich?

2. Which piece does Abed choose?

Definition 4.3.2 (The Lone Divider Method)

1. The divider divides the item into N pieces, which we'll label s_1, s_2, ..., s_N.
2. Each of the choosers will separately list which pieces they consider to be a fair share. This is called their bid.
3. The lists are examined. There are two possibilities:
   (a) If it is possible to assign each player a piece that they bid on, then do so, and the divider gets the remaining piece.
   (b) If this is not possible, then the divider receives an uncontested piece, and the remaining $N-1$ parties repeat the entire procedure. If there are only 2 players left, they may use the Divider Chooser method.

Example 4.3.3 Consider the following example. Who is the divider? Is there a fair division for this example? Why does the Lone Divider method guarantee a fair division in this example?

	Piece 1	Piece 2	Piece 3
Chad	40%	30 %	30 %
Brody	45%	30%	25%
Mary	33.3%	33.3%	33.3%

Example 4.3.4 Consider the following table, which contains information about how much each party values each piece created by the divider. Describe a fair division for this situation.

	Piece 1	Piece 2	Piece 3	Piece 4
Abby	15%	30 %	20%	35%
Brian	30%	35%	10%	25%
Chris	20%	45%	20%	15%
Dorian	25%	25%	25%	25%

Example 4.3.5 Consider the following table, which contains information about how much each party values each piece created by the divider. Describe a fair division for this situation, if one exists. If one does not exist, decide which piece the divider should get and determine what percentage each of the other two players is guaranteed.

	Piece 1	Piece 2	Piece 3
Chooser 1	40%	30 %	30 %
Chooser 2	40%	35%	25%
Divider	33.3%	33.3%	33.3%

Let's take a look at other ways of sharing between 3 or more people!

Definition 4.3.6 (The Last Diminisher)

1. The first person cuts a slice they value as a fair share.
2. The second person examines the piece
   (a) If they think it is worth less than a fair share, they then pass on the piece unchanged.
   (b) If they think the piece is worth more than a fair share, they trim off the excess and lay claim to the piece. The trimmings are added back into the to-be-divided pile
3. Each remaining person, in turn, can either pass or trim the piece
4. After the last person has made their decision, the last person to trim the slice receives it. If no one has modified the slice, then the person who cut it receives it.
5. Whoever receives the piece leaves with their piece and the process repeats with the remaining people. Continue until only 2 people remain; they can divide what is left with the divider-chooser method.

Can you explain why this method results in a fair division?

Definition 4.3.7 (The Moving Knife Method) This method is described with the terminology of a cake and a knife, but can be applied to many different situations.

1. A referee starts moving a knife from left to right across the cake.

2. s soon as any player feels the piece to the left of the knife is worth a fair share, they shout "STOP." The referee then cuts the cake at the current knife position and the player who called stop gets the piece to the left of the knife.

3. This procedure continues until there is only one player left. The player left gets the remaining cake.

Can you explain why this method results in a fair division?
Finally, every method we have discussed so far requires splitting up the item to be shared. What could we do if splitting the item is not allowed and there are 2 parties? What if there are more parties?